

Programming 1

Orga

- Everyone has a working development environment?
- Everyone has a solid understanding of variables, types and collections?
- Any questions so far?

Calendar

6./7.10.	Intro, Orga, Hello, World!	8./9.12.	lectures
13./14.10.	Variables, basics	15./16.12.	mid-term
20.21./10.	Collections, Lists, ...	22./23.12.	lectures
27./28.10.	Conditions and Loops	12./13.1.	lectures
3./4.11.	Functions and Modules	18./19.1.	recap
10./11.11.	Classes and Methods	from 26.1.	exams...
17./18.11.	UI + TK basics		
24./25.11.	Spaceship		
01./02.12.	Invaders and Collisions		

Conditions, Loops

Conditional Tests

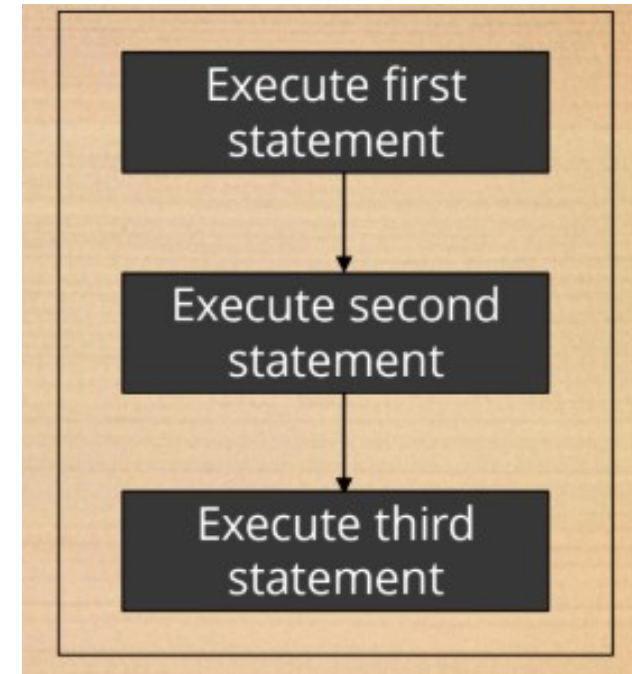
- Checking for Equality:
 - This test compares the current value of a variable to a value of interest
 - `Car == "BMW"` would either return True or False based how it was defined
- Checking for inequality:
 - To determine if two values are not equal
 - `Car != "BMW"` # We use the exclamation mark before the equal to sign to represent not equal to

Conditional tests

- Numerical comparisons:
 - Very straightforward as the title suggests we test numerical values
 - `Age == 18`
 - Many other mathematical operations can be included such as `>`, `<`, `>=`, `<=`
- Checking multiple conditions: This can be done through merging any of the above with the keywords (“and”, “or”)
 - `Age1 >= 18 or Age1 >= 21`
 - `Age >= 18 and Height >= 182`

Sequence

- Sequence as the name implies, is the execution of statements or functions one after the other



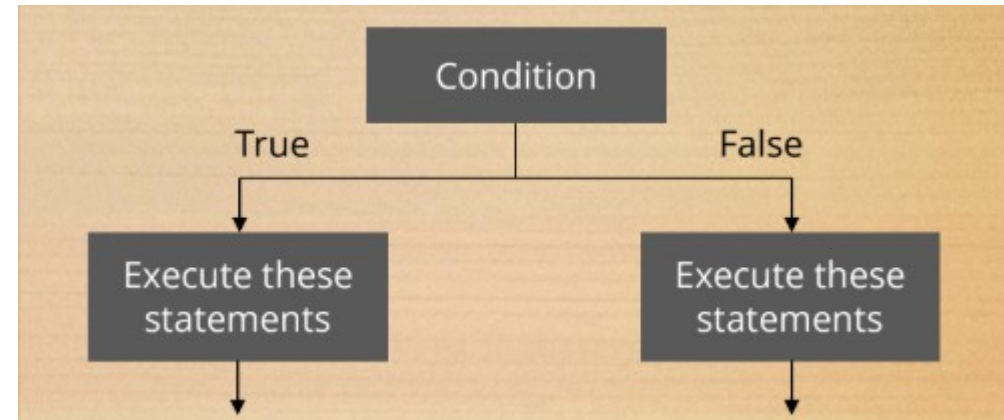
Sequence : Area and circumference of a rectangle

- Steps:
 - Length is asked from the user
 - Width is asked from the user
 - Circumference is calculated using $2 * (\text{Length} + \text{Width})$
 - Area is calculated using $\text{Length} * \text{Width}$
 - Both results are displayed

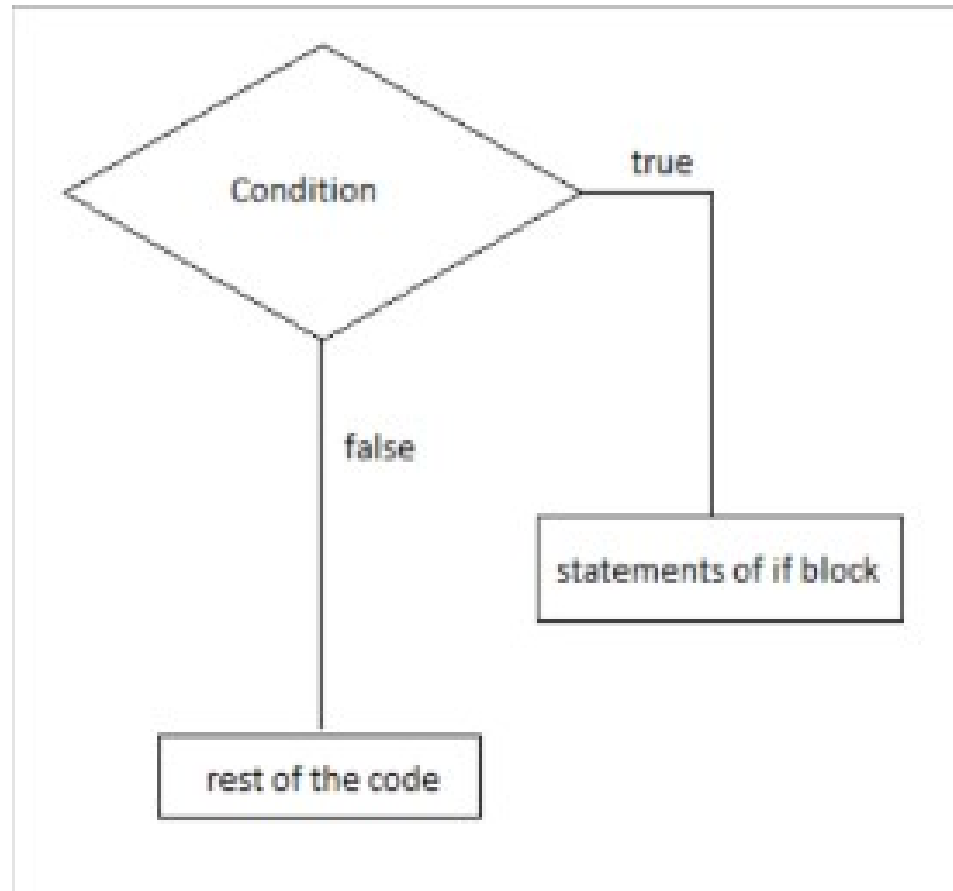
```
length = float(input("Enter the length of a Rectangle: "))
Enter the length of a Rectangle: 10
width = float(input("Enter the width of a Rectangle: "))
Enter the width of a Rectangle: 2.5
C = 2*(length+width)
A = length*width
print("Circumference: ", C)
Circumference: 25.0
print("Area: ", A)
Area: 25.0
```

Selection

- Selection is used to execute only a particular set of statements if a condition is satisfied
- The program may have many paths from the starting point to the end point
- The path is chosen according to the condition satisfied



If Statement



If Statement

If Statement :

if <test expression>:

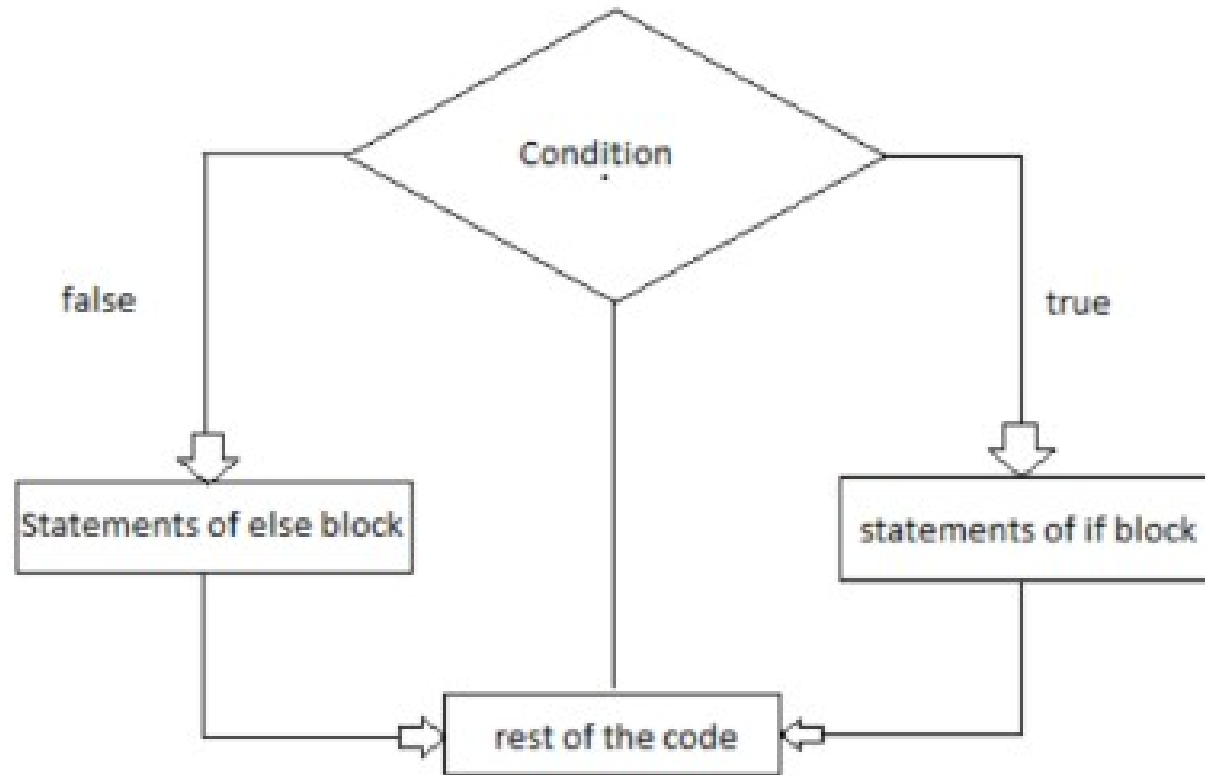
statement(s)

else:

statements(s)

- These statement(s) are executed if the test expression is true
- These statement(s) are executed if the test expression is false

If....else Statement



If...else statements

If else statements

if <test expression>:

statement(s)

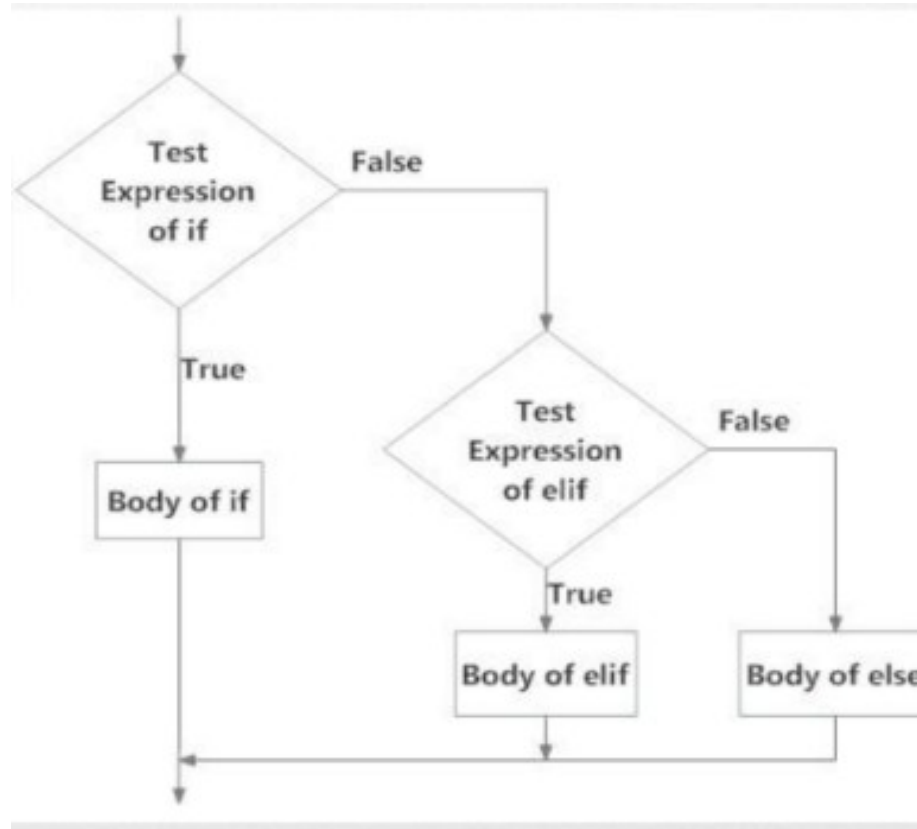
else:

statement(s)

statement(s)

- statement(s) are executed if test expression is true
- statement(s) are executed if test expression is false
- statement(s) are executed after the test expression is checked

If ...elif...else statements



If...elif...else statements

if <test expression>:

statement(s)

elif <test expression 2>:

statement(s)

else:

statement(s)

- These statement(s) are executed if the test expression is true
- These statement(s) are executed if the test expression is false and test expression 2 is true
- These statement(s) are executed if and only if both test expression and test expression 2 are false

Practice programs

- Implement the following algorithm to find the profit/loss based on income and cost.

```
Step 1: Start
Step 2: Read/Input income
Step 3: Read/Input cost
Step 4 : Check if income >=
cost then
    4.1 Calculate profit as
income – cost
    4.2 Display/Print profit
Step 5: Else
    5.1 Calculate loss as cost
– income
    5.2 Display/Print loss
Step 6: End
```

Practice Program

- Implement the algorithm to check the Greater of 2 numbers

Practice Program

- Implement the algorithm to check the Greater of 2 numbers
- Implement an algorithm to check if a number is positive, negative or zero. (*Use a nested if-else statement)
 - A nested if else in simple words is an if else block inside another if else statement

Practice Program

- Implement the algorithm to check the Greater of 2 numbers
- Implement an algorithm to check if a number is positive, negative or zero. (*Use a nested if-else statement)
 - A nested if else in simple words is an if else block inside another if else statement
- Write a program to check if a person is allowed to ride a roller coaster. This means that they need to be older than 10 years old and be taller than 140cm.

Iteration

- Iteration is used to execute a particular set of statements repeatedly until a condition is satisfied. There are three ways to create iteration:
 - Counting how many times the statements are to be executed
 - Repeatedly executing the statements until a condition is true
 - Repeatedly executing the statements while a condition is true

range() Function

- It has the following format:
 - Range (**start**, **stop**, **step**)
 - The sequence includes **start** value and stops at **stop – 1** value
 - The default values are : **start – 0** , **step – 1**
 - All arguments must be integers. Float numbers or any other type can't be used.
 - The step value must NOT be 0. If a step is 0 Python raises a ValueError exception.

RANGE function

- Three variants:
 - `range(stop)` : `range(5)` – 0,1,2,3,4
 - `range(start,stop)`: `range(2,5)` – 2,3,4
 - `range(start,stop,step)`: `range(5,50,10)` – 5,15,25,35,45

Membership Operators

- The membership operators (in , not in) in Python are used to test whether a value is found within a sequence of data.

```
10 in [1,2,10]
```

```
True
```

```
20 in [1,2,3]
```

```
False
```

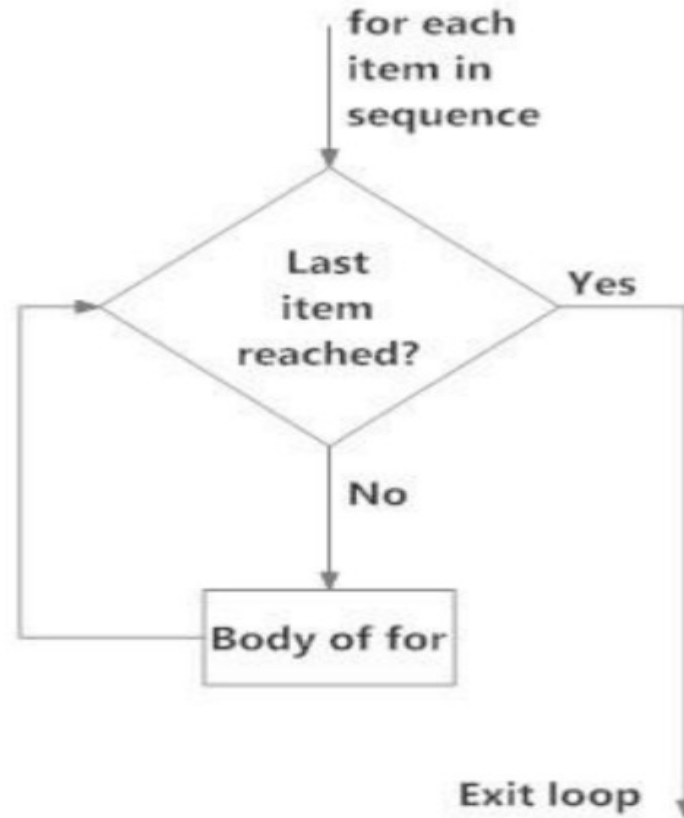
```
20 in range(10)
```

```
False
```

```
3 in range(10)
```

```
True
```

For loops



For loops

- A for loop executes a block of code which you want to repeat a fixed number of times. The range function is used here.

Syntax:

for <iterator> in <sequence>:

statement(s)

statement(s)

- These statement(s) are executed the number of times mentioned in the sequence
- These statement(s) are executed after the for loop is done executing

For loop

```
for i in range(1,4):  
    print("inside statement number ",i)
```

```
inside statement number 1  
inside statement number 2  
inside statement number 3
```

Jump statements

- Used to alter the flow of the loop – either terminate the loop or skip some statements within a loop
 - Break
 - Continue
 - Pass

Pass

- The pass statement is a null statement
- It is useful in those instances where empty code is not allowed, like in loops, function definitions, class definitions or in if statements

Pass

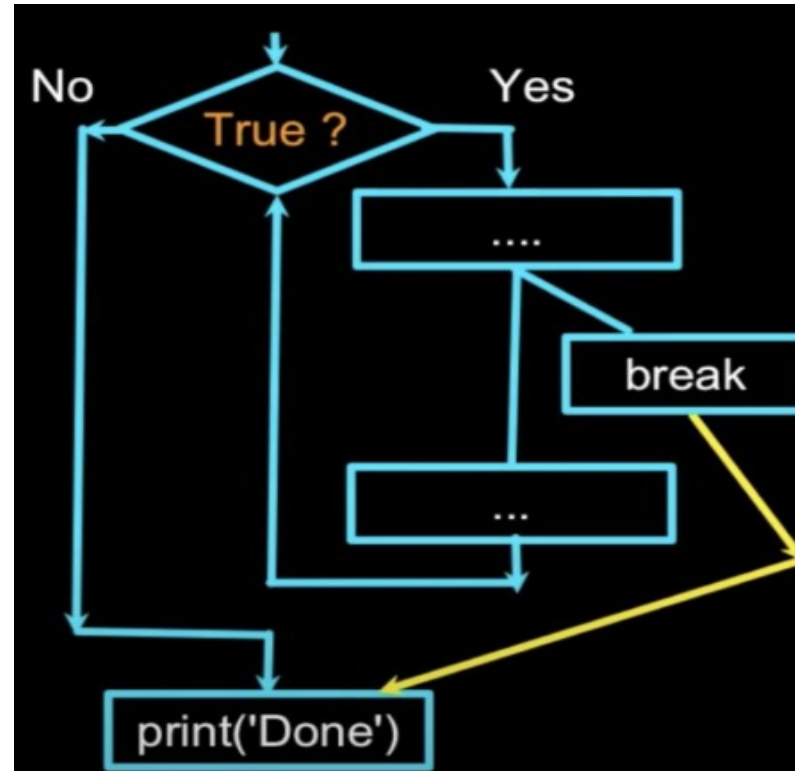
```
for n in range(5):  
    if n==3:  
        pass  
    else:  
        print(n, "Else block")  
print('Done with the loop')
```

Output:

```
0 Else block  
1 Else block  
2 Else block  
4 Else block  
Done with the loop
```

Break

- A break statement is used to terminate a loop



Break

```
for n in range(10):
    if n==3:
        print("Before break")
        break
    print("after break")
print(n)
```

Output:

0

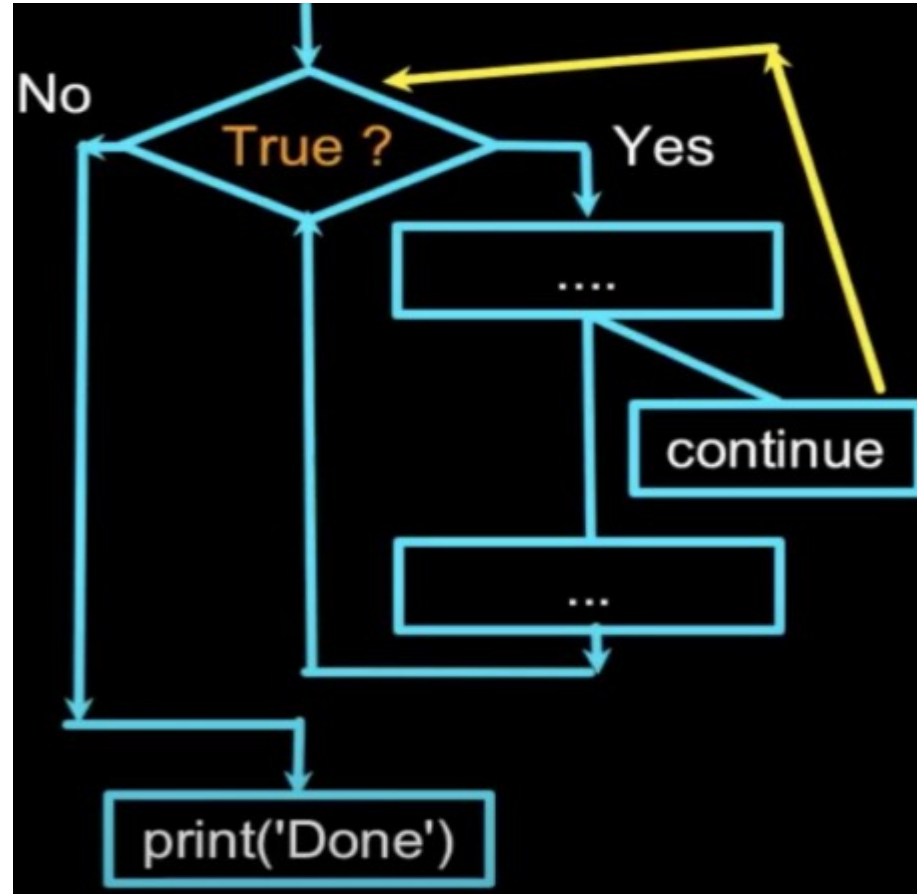
1

2

Before break

Continue

- A continue statement is used to skip all the remaining statements in the loop and move controls back to the top of the loop.



Continue

```
for n in range(6):  
    if n==3:  
        print("before continue")  
        continue  
        print("after continue")  
    print(n)
```

Output:

```
0  
1  
2  
before continue  
4  
5
```

Nested for loop

- A for loop inside another for loop is known as a nested for loop.
- They are usually used to print a 2D matrix/ an $m \times n$ matrix, patterns.
- The following link can be used to visualize how a nested for loop works
- <https://pythontutor.com/python-compiler.html#mode=edit>
- The code snippet is on the next slide

Nested for loop

```
for i in range(5):
    for j in range(1,3):
        print(i, ', ', j)
```

```
0, 1
0, 2
1, 1
1, 2
2, 1
2, 2
3, 1
3, 2
4, 1
4, 2
```

End of lectures

Any questions??

Practice programs

- Using a for loop print the multiplication table for the following numbers : 13,7,19
- Display the sum of the first 10 natural numbers
- Display the squares of the numbers from 11 to 20. Output should look like: 11 – 121, 12 – 144, 13 – 169.....
- Display the characters A – Z in reverse order. [Use the internet to research the ascii values of the characters]

While loop

- Executes a block of code as long as test expression is true
- Syntax:

```
while
<expression>:
    statement(s)
```

- These statement(s) are executed as long as the expression is true

While loop – steps to remember

- Initialize the iterator outside the while loop
- Test Expression with while
- Increment / Decrement the iterator within the while loop

```
i = 0 #Iterator outside the while loop
while i<3: #Test Expression with while
    print(i)
    i+=1 #increment the iterator within the while loop
```

```
0
1
2
```

Practice Programs

- Write a program to generate a number between 50 and 100 using the random module and ask the user to guess the number, giving hints along the way
- Write a program to input some numbers repeatedly and print their sum. The program ends when the user enters a number smaller than 0